

Predicting Clinical Health Risk from Neurobehavioral & Lifestyle Indicators

1. Project Definition

1.1 Problem Statement

Most chronic diseases are only caught once something has already gone wrong biologically. Doctors usually assess risk using separate tools for different areas. A cardiovascular calculator looks at lipid panels. A mental health screener looks at behavioral questionnaires. A metabolic checklist looks at things like glucose and waist circumference, but these tools usually look at one area at a time. Because of that, a patient with slightly high CRP, poor sleep, high stress, and borderline LDL might not get flagged anywhere, even though the combination is concerning. This project builds a machine learning model that combines behavioral, neurological, and biomarker data to predict whether a patient is Low, Medium, or High risk. The dataset did not include a target variable, so the risk labels had to be engineered using clinical thresholds. The main focus of the project was feature engineering across all three domains.

1.2 Importance

Many AI models in healthcare focus mostly on imaging or clinical lab data and ignore behavioral factors like stress and sleep (Rajpurkar et al.). This project tries to combine both. Lifestyle features are easier to collect than blood work and can still provide meaningful health signals. Research also shows that chronic stress is linked to cardiovascular disease and increased CRP levels (Kivimaki and Steptoe). That relationship supports combining behavioral and biomarker data in the same model. Most existing studies use either clinical or behavioral data, but not both. This project combines five feature groups in one pipeline.

1.3 Connection to Course Material

I used several of the course concepts in one pipeline. In the ETL step, I used RegEx to clean and standardize the Gender column so things like "male", "FEMALE", or extra spaces wouldn't cause issues later. I also normalized the data into three PostgreSQL tables using PRIMARY KEY, FOREIGN KEY, and CHECK constraints, then joined them back together for modeling. I applied K-Means as a preprocessing step and added the cluster label as a feature, which connects the unsupervised and supervised parts of the project. For modeling, I trained Random Forest and XGBoost using 5-fold stratified cross-validation, and evaluated them with learning curves and SHAP to understand what the models were doing.

2. Related Work

I first considered logistic regression, but I did not use it because this problem depends heavily on feature interactions. LDL alone is not very informative, but LDL combined with low HDL and high triglycerides is much more meaningful. Logistic regression would require manually defining those interactions. Tree-based models handle them automatically, so I used Random Forest as a baseline and XGBoost as the main model.

For feature engineering, I used established clinical formulas. I used the HOMA-IR formula, $(\text{Fasting Glucose} \times \text{Insulin}) / 405$, as a proxy for insulin resistance (Matthews et al.). I also built a Metabolic Risk Index using ATP III clinical guidelines, which define the importance of different biomarkers in cardiovascular risk. For interpretation, I used SHAP instead of only feature importance. Feature

importance measures how often a feature is used in model splits, while SHAP measures how much a feature changes predictions.

3. Progress and contribution

3.1 Data Utilization

The dataset used is the NeuroBehavior Clinical Health Risk Sample Dataset from Kaggle (Sumon). It has 1,000 records across five groups which are demographics (age, gender), lifestyle factors (sleep hours, stress level, screen time, physical activity, meditation, social interaction), cognitive and emotional indicators (rumination score, focus level, mood score), hormonal proxies (cortisol risk, dopamine balance, oxytocin score), and clinical markers (CRP, ESR, fasting glucose, insulin, lipid profile, liver enzymes). The dataset is synthetic, meaning the features were generated independently and have almost no real correlations. That is the main trade-off and I document it in section 3.6.

Initial exploration showed a mean Fasting Glucose of 109.92 mg/dL (pre-diabetic range) and mean LDL of 128.30 mg/dL (borderline high), suggesting the dataset skews toward a higher-risk population. This is consistent with the class distribution after engineering the target label where 69.7% of patients ended up in the Medium risk category.

3.2 Cleaning the Data

Preparing the Data:

Since the dataset came from Kaggle and was described as synthetic, I initially assumed it would already be clean. That was mostly true, but I still ran into issues. One problem came from outlier detection. I originally ran IQR detection on all numeric columns using `select_dtypes(include=np.number)`. Later, after adding engineered features like `is_outlier_flagged`, the pipeline started detecting its own metadata columns as outliers. A mostly binary column technically contains outliers under IQR rules. I fixed this by creating a `raw_cols` list containing only the original 22 clinical columns and running outlier detection only on those.

Cleaning the Data:

My main goal when cleaning the data was to validate clinical plausibility rather than just fix missing values, since the dataset had none. The ETL pipeline runs ten distinct checks:

I started by checking the basics like row count, duplicates, and missing values. The dataset had 1,000 rows, no duplicates, and no missing values. After that, I validated the Gender column using a RegEx pattern `^(Male|Female)$` with ignore case. This was to catch possible real-world entries like "male", "FEMALE", "M", "F", or values with extra spaces. The dataset itself was clean, but I added this step because if the data is updated or combined with another source later, inconsistent categorical values could break encoding further down the pipeline.

```
VALID_GENDER_PATTERN = re.compile(r'^(Male|Female)$', re.IGNORECASE)
```

I also ran clinical range checks on key columns using reasonable bounds. For example, CRP was checked between 0 and 10 mg/L, fasting glucose between 50 and 400 mg/dL, and sleep hours between 4 and 12. If any values were outside these ranges, I planned to clamp them instead of dropping rows so the data would still be usable. In this case, nothing needed to be corrected, which confirmed the data was already within valid ranges.

For outliers, I applied both IQR and Z-score methods, but only on the original clinical columns, not the engineered ones. No outliers were detected. I also added a few domain-specific checks. All Likert-scale columns were confirmed to stay within their defined ranges, screen time plus sleep did not exceed 24 hours, and HDL did not exceed LDL. None of these checks flagged issues, but including them was important because if the data changes later errors will be caught immediately.

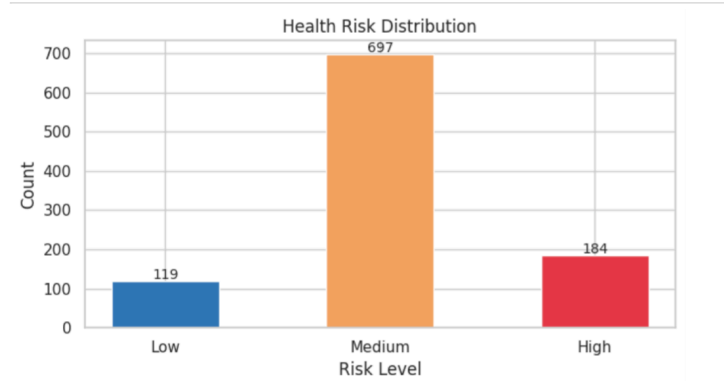


Figure 1: Health risk label distribution after target engineering (69.7% Medium, 18.4% High, 11.9% Low)

3.3 Models, Techniques, and Algorithms

Feature Engineering

I first trained Random Forest using only raw features. It reached about 73.4% CV accuracy. But clinically, biomarkers should not be interpreted separately. LDL alone is less informative than LDL combined with HDL, triglycerides, glucose, and CRP. To address this, I created a Metabolic Risk Index that combines LDL, HDL, triglycerides, fasting glucose, and CRP. Each value was normalized by its clinical threshold. For example, LDL divided by 130 means a patient at the threshold scores 1, below scores less, and above scores more. I capped values at 2 to prevent extreme effects. The weights were based on clinical importance from ATP III criteria: LDL 25%, HDL 20%, triglycerides 20%, glucose 20%, CRP 15%. After adding this feature, CV accuracy increased to 76.5%.

I then built six more composite features using the same approach. These included a Psychological Burden Score, a Neurochemical Balance Index, a Lifestyle Quality Score, an Inflammation Index, a Sleep Quality Flag, and a HOMA-IR proxy using $(\text{Fasting Glucose} \times \text{Insulin}) / 405$. Not all of them helped. The Neurochemical Balance and Lifestyle Quality features had low SHAP values and overlapping distributions across all classes, meaning they were not useful for separating risk groups. I kept them anyway because removing them did not change performance much, and they still represent an attempt to model those domains. This also says something about the dataset. The neurochemical features are not predictive here, likely because they were generated independently from the biomarkers.

Target Variable

Creating the target variable was one of the harder parts of the project. I went through a few versions. The first version used only a few biomarker thresholds. If one was elevated, the patient was Medium risk. If two or more were elevated, they were High. This gave a decent class split, but the model learned it too easily. The confusion matrix was almost perfect, which was a red flag. The label was basically just a direct function of the same features used for training.

The second version added behavioral features, but I weighted them too heavily. Patients with high stress and poor sleep were being labeled High risk even if their biomarkers were normal. That does not make sense clinically. Behavioral factors matter, but they do not directly indicate current physiological risk.

The final version uses a combination of biomarker and behavioral thresholds but keeps the focus on biomarkers. Each patient gets a score based on whether they exceed certain thresholds. These include CRP, fasting glucose, LDL, HDL, triglycerides, ALT, stress, sleep, rumination, and mood. Scores of 0 to 2 are Low risk, 3 to 5 are Medium, and 6 or more are High. These thresholds are based on clinical guidelines. This produced 11.9% Low, 69.7% Medium, and 18.4% High, which is a realistic distribution for a higher-risk population.

Database Schema

For the database part, I normalized the dataset into three PostgreSQL tables instead of using one large table. A single table would have been simpler, but normalization made more sense here. Separating demographics, behavioral features, and biomarkers keeps the data more organized and avoids mixing unrelated columns. It also makes queries more efficient and flexible. For example, if I only need lab results, I do not have to query behavioral data. This would matter more at scale and also for access control, where different users might only be allowed to see certain types of data.

I added CHECK constraints to each table to enforce the same validation rules used in the ETL step. This makes sure that even if data is inserted directly into the database later, invalid values are still blocked. To use the data for modeling, I joined the tables back together using patient_id:

```
SELECT
  p.patient_id, p.age, p.gender,
  b.stress_level, b.sleep_hours, b.mood_score,
  bio.crp, bio.fasting_glucose, bio.health_risk
FROM patients p
JOIN health_behavioral b ON p.patient_id = b.patient_id
JOIN health_biomarkers bio ON p.patient_id = bio.patient_id
LIMIT 5;
```

```
CREATE TABLE patients (
  patient_id INT PRIMARY KEY,
  age INT CHECK (age >= 0),
  gender VARCHAR(10) CHECK (gender IN ('M', 'F')),
  stress_level INT CHECK (stress_level >= 0),
  sleep_hours INT CHECK (sleep_hours >= 0),
  mood_score INT CHECK (mood_score >= 0),
  crp INT CHECK (crp >= 0),
  fasting_glucose INT CHECK (fasting_glucose >= 0),
  health_risk INT CHECK (health_risk >= 0)
);

CREATE TABLE health_behavioral (
  patient_id INT PRIMARY KEY,
  stress_level INT CHECK (stress_level >= 0),
  sleep_hours INT CHECK (sleep_hours >= 0),
  mood_score INT CHECK (mood_score >= 0),
  rumination INT CHECK (rumination >= 0),
  health_risk INT CHECK (health_risk >= 0)
);

CREATE TABLE health_biomarkers (
  patient_id INT PRIMARY KEY,
  crp INT CHECK (crp >= 0),
  fasting_glucose INT CHECK (fasting_glucose >= 0),
  cholesterol INT CHECK (cholesterol >= 0),
  triglycerides INT CHECK (triglycerides >= 0),
  hdl INT CHECK (hdl >= 0),
  ldh INT CHECK (ldh >= 0),
  alt INT CHECK (alt >= 0),
  health_risk INT CHECK (health_risk >= 0)
);
```

Figure 2: PostgreSQL schema show 3 normalized tables with PRIMARY KEY, FOREIGN KEY, and CHECK constraints matching the ETL clinical validation rules

Clustering

I was not sure if K-Means clustering would help because the dataset did not show strong natural groupings. To test this, I first trained the model without clustering and recorded the baseline metrics. Then I added the K-Means cluster label as a feature and retrained the model. Adding the cluster label slightly improved performance. CV accuracy went from 83.9% to 84.1%, and F1 increased from 0.822 to 0.823. The bigger signal came from feature importance. The cluster label had a relatively high importance score at 0.062, which suggests it was helping the model split the data even if the clusters themselves were not clearly separated.

To choose k , I ran the elbow method and silhouette scores for k from 2 to 9. Both suggested $k=2$ as the most distinct grouping, but that does not match the three-class risk structure. I chose $k=3$ to align with the target variable, even though the silhouette scores were low, around 0.05 to 0.085. The PCA plot shows heavy overlap between clusters, which is expected since the first two components only explain about 16.6% of the variance.

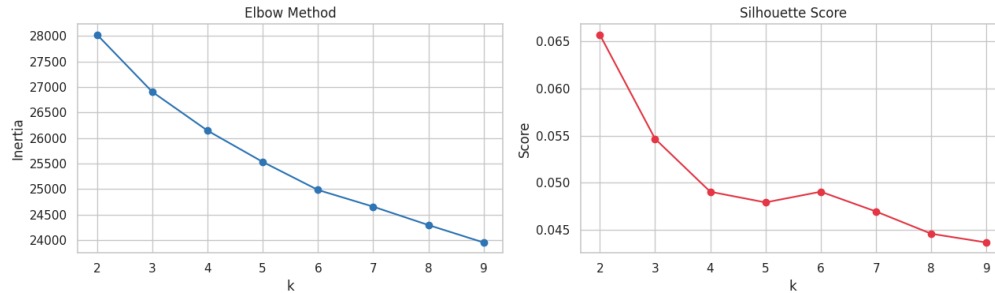


Figure 3: Elbow method and silhouette scores for $k=2$ to $k=9$. Low scores confirm no strongly separated natural clusters. $k=3$ chosen on domain grounds

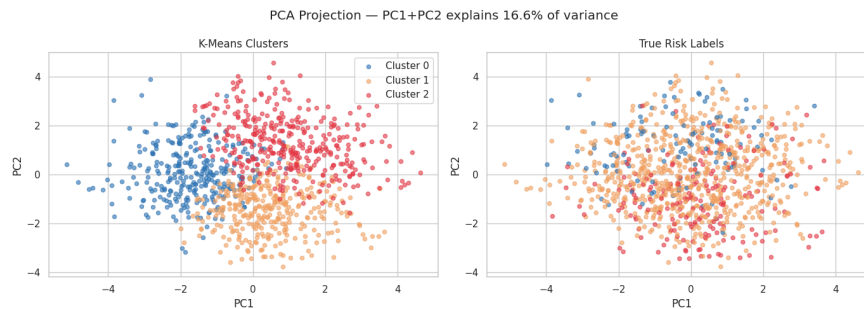


Figure 4: PCA projection. K-Means clusters (left) vs true risk labels (right). Overlap is expected given synthetic feature independence and 16.6% variance explained

Model Training:

I trained Random Forest and XGBoost using 5-fold stratified cross-validation. Stratification was important because of class imbalance. I used `class_weight='balanced'` in both models.

I also tested SMOTE, but it performed worse. Since the dataset is already synthetic, SMOTE created noisy artificial samples instead of improving the signal. The data was split using an 80/20 stratified train-test split with `stratify=y`. The test set remained completely separate and was not used during training or cross-validation. This makes the test accuracy, which was about 83% for XGBoost, a more reliable measure of performance.

```

Train size: 800
Test size: 200

Train:
  High: 147 (18.4%)
  Low: 95 (11.9%)
  Medium: 558 (69.8%)
Test:
  High: 37 (18.5%)
  Low: 24 (12.0%)
  Medium: 139 (69.5%)

```

Figure 5: Stratified 80/20 split. class proportions maintained across training (800 samples) and test (200 samples) sets

3.4 Results

Table 1 summarizes performance for both models:

	CV Accuracy	CV F1	CV ROC-AUC	Train Acc	Gap
Random Forest	0.7825	0.7443	0.8739	1.0	0.2175
XGBoost	0.8112	0.7914	0.8976	1.0	0.1888

Table 1: Model performance comparison using 5-fold cross-validation and a held-out test set ($n=200$)

XGBoost outperformed Random Forest on all metrics. The CV ROC-AUC of 0.8976 shows the model separated the three risk classes fairly well. On the held-out test set, XGBoost correctly classified 166 out of 200 patients. The main problem was overfitting. Both models reached 1.0 training accuracy, meaning they memorized the training data. The gap between training and validation accuracy is about 0.19 for XGBoost, which is pretty large. I tried lowering `max_depth`, adding more regularization, and reducing the number of trees. These changes reduced overfitting slightly, but they also reduced validation performance. The trade-off was not worth it. The bigger issue is probably not model complexity itself, but how the target variable was created. Since the labels were created using some of the same features used for training, the model could partly learn the labeling rule itself. That likely inflates performance and means the results may not generalize as well if there was new unseen data.

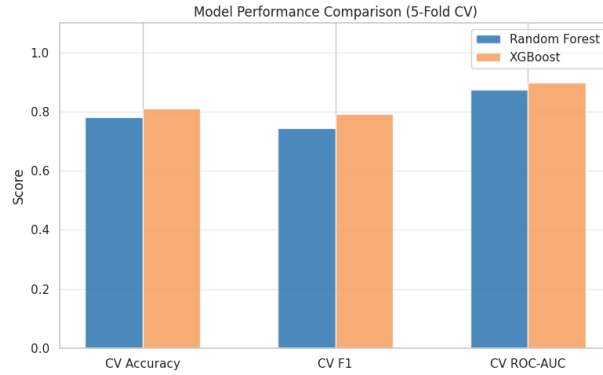


Figure 6: Model comparison across CV accuracy, F1, and ROC-AUC, with XGBoost performing better on all metrics

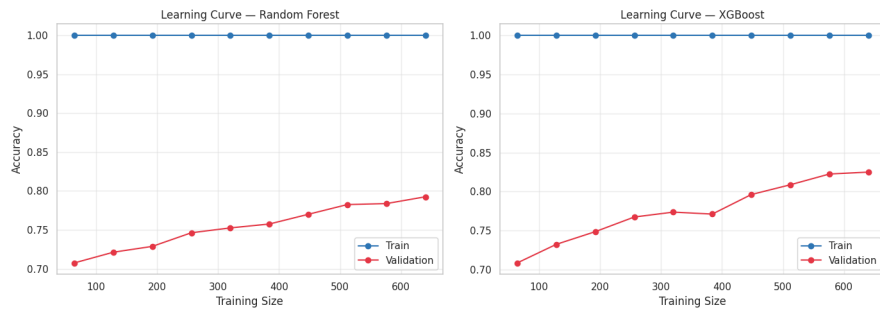


Figure 7: Learning curves showing training accuracy at 1.0 and lower validation accuracy, indicating overfitting

Looking at performance by class shows another issue. XGBoost gets a recall of 0.96 for the Medium class, which sounds impressive at first, but Medium makes up about 70% of the dataset. The weaker results show up in the smaller classes. For High risk, recall is only 0.57, meaning the model misses 16 out of 37 High-risk patients in the test set and predicts them as Medium instead. For Low risk, recall drops to 0.46, so more than half of the Low-risk patients also get pushed into the Medium category. In a real clinical setting, missing High-risk patients would be the most serious mistake the model could make. One positive is that the model almost never confuses High with Low directly, but the tendency to default to Medium whenever it's uncertain is still a major issue. This mostly comes from class imbalance. Even with balanced class weights, the model still sees far more Medium examples during training, so Medium becomes the “safe” prediction whenever confidence is low.

CV Accuracy: 0.7825
CV F1: 0.7443
CV ROC-AUC: 0.8739

	precision	recall	f1-score	support
High	0.89	0.46	0.61	37
Low	1.00	0.33	0.50	24
Medium	0.79	0.99	0.88	139
accuracy			0.81	200
macro avg	0.90	0.59	0.66	200
weighted avg	0.84	0.81	0.78	200

Figure 8: Random Forest classification report showing strong Medium recall and weaker performance on High and Low classes

CV Accuracy: 0.8112					
CV F1: 0.7914					
CV ROC-AUC: 0.8976					
	precision	recall	f1-score	support	
High	0.83	0.54	0.66	37	
Low	0.92	0.50	0.65	24	
Medium	0.82	0.96	0.89	139	
accuracy			0.83	200	
macro avg	0.86	0.67	0.73	200	
weighted avg	0.84	0.83	0.82	200	

Figure 9: XGBoost classification report showing better recall for Low and High compared to Random Forest

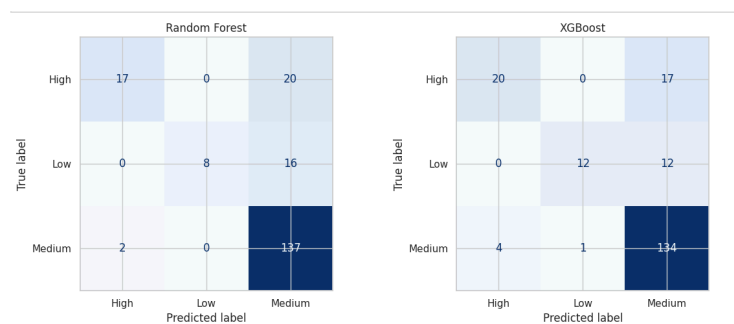


Figure 10: Confusion matrices on the test set ($n=200$) showing most predictions fall into Medium, with errors mainly between Medium and the other classes

3.5 Data Visualization

The correlation heatmap ended up being one of the most useful outputs. I expected to see some relationships, like LDL and triglycerides increasing together, stress and rumination being linked, or poor sleep affecting cortisol. Instead, almost all correlations were below 0.10. The highest in the entire matrix was around 0.10. This was a clear sign that the data is synthetic. The features are mostly independent, which is not how real clinical data behaves. Real clinical data would usually have correlations around 0.3–0.6 between related features. Without those relationships, the model can't really learn patterns like “high LDL combined with low HDL and high stress” being a metabolic syndrome profile. It mostly just learns separate threshold rules, like whether a value is above some cutoff or not. That's more of a limitation of the synthetic data than the model itself.

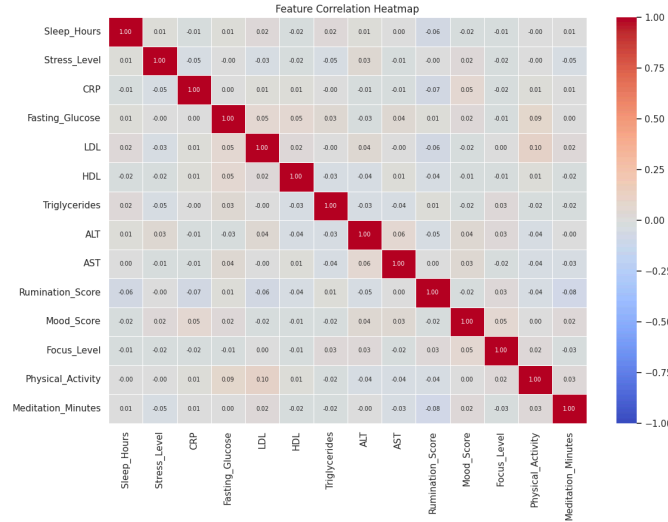


Figure 11: Correlation heatmap showing all pairwise correlations below 0.10, confirming that the features are mostly independent. In real clinical data, related features would usually show moderate correlations.

The biomarker distributions help validate the target variable. Each biomarker shifts in the expected direction across risk groups. CRP, fasting glucose, LDL, triglycerides, and ALT increase from low to high risk, while HDL decreases, which is correct since it's the protective cholesterol and lower HDL means higher risk. If this pattern was not present, it would suggest a problem with how the risk label was defined. The fact that every biomarker moves as expected is confirmation that the ten threshold flags captured genuine clinical risk rather than arbitrary cutoffs.

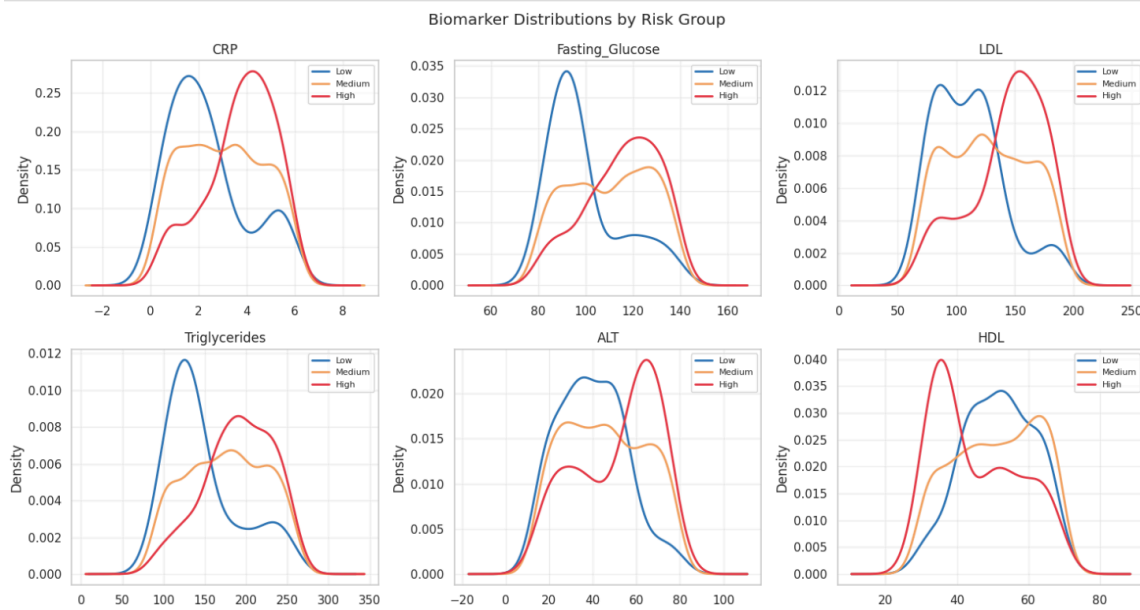


Figure 12: Biomarker distributions by risk group showing clear separation that matches clinical expectations.

The composite features were mixed. The Metabolic Risk Index shows clear separation across risk groups. The High-risk distribution is shifted right compared to Low and Medium, which supports how it was built. Combining LDL, HDL, triglycerides, glucose, and CRP into one weighted index gave the model a stronger signal than any single biomarker by itself. But the Neurochemical Balance and Lifestyle Quality distributions show almost complete overlap across all three groups. The curves are basically sitting on top of each other, so those features are not really helping the model separate the risk classes. This goes back to the heatmap. Since the features were generated independently with almost no real correlations, combining them into composites doesn't create much meaningful signal. In real clinical data, things like dopamine balance or cortisol risk would probably correlate with stress, sleep, and other lifestyle factors, so a neurochemical composite could actually become informative. Here, those relationships just don't exist in the data.

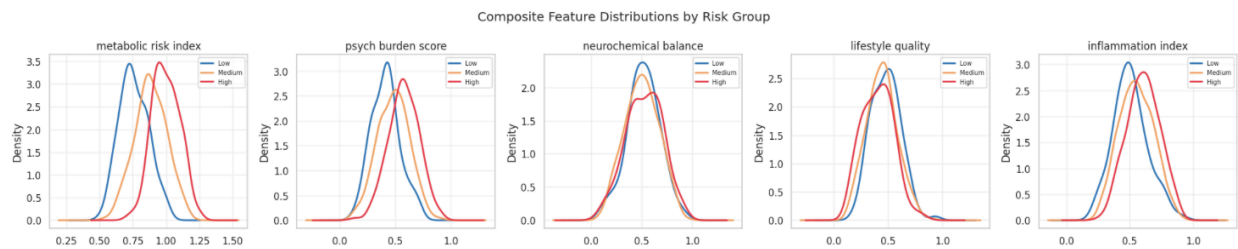


Figure 13: Composite feature distributions showing strong separation for the Metabolic Risk Index and little to no separation for the other composite features.

Feature importance and SHAP give a clearer view of what the model is using.

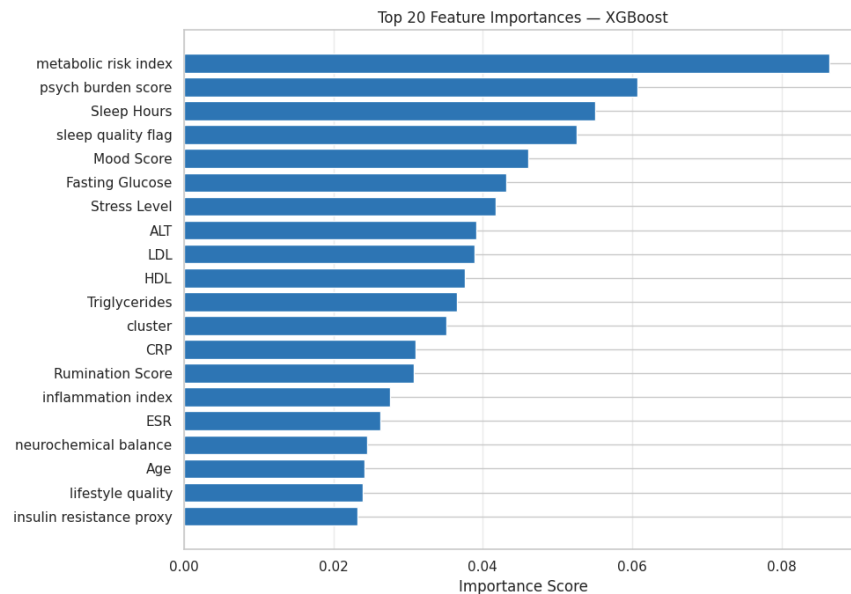


Figure 14: XGBoost feature importance showing Metabolic Risk Index as the top feature

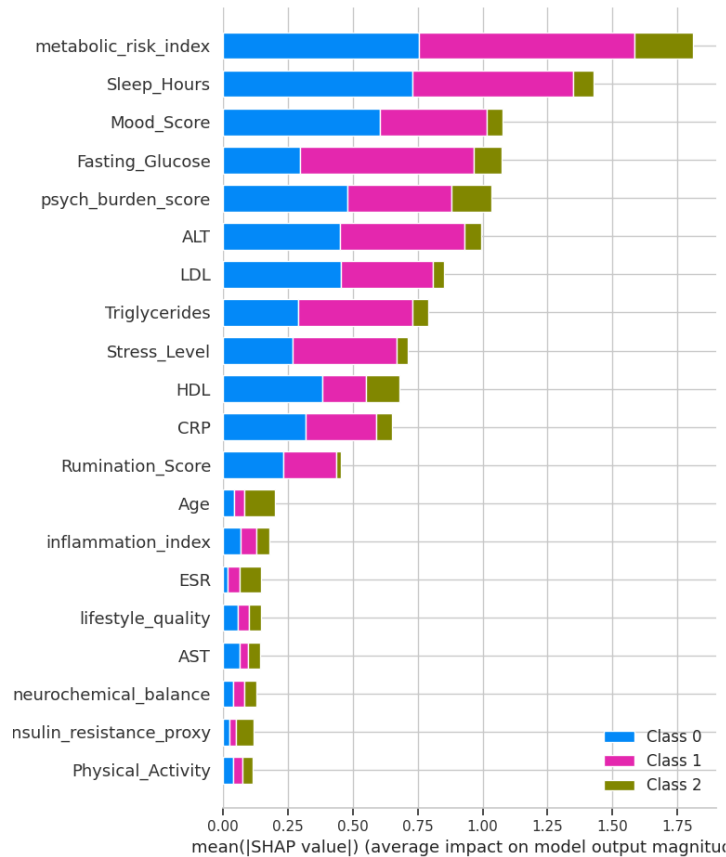


Figure 15: SHAP summary showing that metabolic and some behavioral features drive predictions, while neurochemical and lifestyle composites contribute very little.

One interesting result is that Sleep_Hours ranked third in SHAP and also ranked high in feature importance, above biomarkers like LDL and triglycerides. Part of that is because sleep below 6 hours is one of the threshold flags, but it also makes sense biologically since sleep affects stress, inflammation, and metabolic health at the same time. The Sleep Quality Flag also ranked surprisingly high even though it's just a binary feature showing whether sleep falls in the 7–9 hour range. It ended up being more useful than some raw continuous biomarkers, which is a good example of a simple engineered feature outperforming more detailed variables. The Psychological Burden Score ranked second in feature importance, which was interesting because the neurochemical composites performed pretty poorly overall. Unlike those features, the Psychological Burden Score combines stress, rumination, mood, and focus, and all of those variables also had threshold flags in the target variable. Because of that, the composite captured more signal than any single behavioral feature on its own. The cluster feature also showed an interesting pattern. It had a relatively high feature importance score at 0.062 but ranked near the bottom in SHAP. Feature importance measures how often a feature gets used in splits, while SHAP measures how much it actually changes the prediction. So the cluster feature is helping organize the feature space, but each individual split has only a small impact on the output. That's pretty much what you'd expect from an unsupervised preprocessing feature, especially with low silhouette scores. The neurochemical and lifestyle composites were near the bottom in both plots, which matches the overlapping distributions from earlier. Keeping them in the model didn't really hurt performance, but they also were not contributing much.

3.6 Advantages and Limitations

The biggest strength of this project is probably the feature engineering. Instead of treating the 22 features as separate variables, I grouped them into clinical domains and built composite features like metabolic risk, psychological burden, inflammation, and lifestyle quality. These ended up being more useful than the raw features, especially the Metabolic Risk Index, which consistently ranked highest. This shows that combining related variables worked better than treating everything independently.

Most of the limitations are connected to each other. Since the dataset is synthetic, the features were generated independently with almost no real correlations. Because of that, the model is mostly learning isolated threshold rules instead of realistic clinical patterns, which is exactly what the heatmap showed earlier. The target variable circularity makes this worse. Since the risk labels were created from some of the same features used for training, the model can partly learn the labeling rule itself, which is probably why both models reached 1.0 training accuracy. Class imbalance is also still a major issue even with balanced class weights. XGBoost only reached 0.57 recall for High-risk patients, meaning it missed almost half of them by predicting Medium instead. In a real clinical setting, that level of missed High-risk cases would not be acceptable. The neurochemical and lifestyle composite features also ended up contributing very little. They didn't improve discrimination between the risk groups, which is honestly a useful negative result to document because it shows the limits of what this kind of synthetic data can realistically support.

4. Changes After Proposal

4.1 Differences from Proposal

The final project ended up being more complex than what I originally planned. I added K-Means clustering after noticing the classifier struggled in a high-dimensional space and tested whether a cluster label would help. It did, so I kept it. I also added cross-validation, learning curves, and SHAP analysis, which were not part of the original plan. The biggest change was feature engineering. I originally planned to use the raw columns, but building composite features based on clinical ideas ended up being the most important part of the project.

4.2 Bottlenecks and Challenges

The biggest challenge was defining the target variable. I went through multiple versions before settling on one that made sense. The first version was too simple, and the model learned it almost perfectly, which showed that the label was basically just a direct function of the features. Class imbalance was another issue. I tried class weights, SMOTE, and threshold adjustments. SMOTE made performance worse, likely because the data is synthetic. Threshold tuning improved High-risk recall but hurt precision, so I stayed with class weighting. I also ran into an issue with outlier detection. After adding new columns, the model started flagging those columns as outliers. The problem was that I was applying outlier detection too broadly. The fix was to only run it on the original raw features. Clustering was another decision point. The metrics suggested weak clusters, but removing clustering felt like the wrong choice. Keeping it ended up helping the model, even if the clusters themselves were not clearly separated.

5. Future Directions

The most important next step is testing this pipeline on real clinical data. The current dataset is synthetic, so it does not show whether the model would work in practice. For class imbalance, a better approach would be to use different costs for different errors. Missing a High-risk patient should matter more than misclassifying a Medium-risk one. Another improvement would be using time-based data instead of a single snapshot. Health risk is based on trends over time, not just one measurement. Tracking changes in biomarkers, sleep, and stress would make the model more useful.

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